

HYPOTHESIS TESTING: QUESTIONS AND ANSWERS

Question 1: Null Hypothesis (H_0)

Ans:-The null hypothesis (H_0) is a statement that there is no effect or no difference, and it serves as the default or starting assumption in hypothesis testing. It is important because it provides a baseline against which the alternative hypothesis (H_1) is tested. Rejecting the null hypothesis suggests that there is sufficient evidence to support the alternative hypothesis. **Example:**

H_0 : Average complaint resolution time = 10 minutes

Question 2: Significance Level (α)

Ans:-The significance level (α) represents the probability of rejecting the null hypothesis when it is actually true (Type I error). Common values are 0.05, 0.01, and 0.10. A smaller α reduces the chance of a Type I error but increases the chance of a Type II error. **Common values:**

$\alpha = 0.05$ (5%), 0.01 (1%)

Question 3: Type I and Type II Errors

Ans:-Type I Error (False Positive): Rejecting the null hypothesis when it is true.

Type II Error (False Negative): Failing to reject the null hypothesis when it is false.

Question 4: One-tailed vs Two-tailed Tests

Ans:-One-tailed Test: Tests for the possibility of the relationship in one direction.

Example: Testing if a new drug is more effective than the current one.

Two-tailed Test: Tests for the possibility of the relationship in both directions. Example:

Testing if a new teaching method has a different effect (better or worse) than the traditional method. Example:

$H_0: \mu = 50$

$H_1: \mu > 50$

Two-tailed test:

Tests effect in **both directions**

Example:

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

Question 5: Test Company Claim (Mean = 10)

Ans:- Given:

- $\mu_0 = 10$
- $\bar{x} = 12$
- $s = 3$
- $n = 9$
- $\alpha = 0.05$

Step 1: Hypotheses

$$H_0: \mu = 10$$

$$H_1: \mu \neq 10$$

Step 2: Test statistic (t-test)

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$
$$t = \frac{12 - 10}{3/\sqrt{9}} = \frac{2}{1} = 2$$

Step 3: Critical value

$$df = 8 \rightarrow t_{0.025} = 2.306$$

Decision:

$$|t| < 2.306 \rightarrow \text{Fail to reject } H_0$$

Conclusion:

There is **not enough evidence** to reject the company's claim.

Given: sample mean = 12, population mean = 10, $s = 3$, $n = 9$, $\alpha = 0.05$

t-statistic = 2.0000

Degrees of freedom = 8

p-value = 0.0805

Conclusion: Fail to reject H_0 (at $\alpha = 0.05$)

Question 6: Z-test vs t-test

Ans:- Use Z-test when the population standard deviation is known and the sample size is large ($n \geq 30$).

Use t-test when the population standard deviation is unknown and/or the sample size is small ($n < 30$).

Question 7: Paired t-test for Productivity

Ans:- Test: Paired t-test

Hypotheses:

$$H_0: \mu_a - \mu_\beta = 0$$

$$H_1: \mu_a - \mu_\beta > 0$$

Formula:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Result:

Calculated $t >$ critical t

Conclusion:

Reject $H_0 \rightarrow$ Training improved productivity

t-statistic = 4.1079

p-value = 0.0093

Conclusion: Reject H_0 (at $\alpha = 0.05$)

Question 8: Chi-square Test for Independence

Ans:- Hypotheses:

H_0 : Product preference is independent of gender

H_1 : Product preference is dependent on gender

Expected frequency formula:

$$E = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

Chi-square formula:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Decision:

χ^2 (calculated) > χ^2 (critical)

Conclusion:

Reject $H_0 \rightarrow$ Product preference depends on gender

Chi-square statistic = 15.0417

Degrees of freedom = 1

p-value = 0.0001

Conclusion: Reject H_0 (at $\alpha = 0.05$)